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Elsevier Journal 165 (1999) 1–15

EPSL

Paleomagnetic study of middle Cretaceous rocks from Yunlong, western Yunnan, China: evidence of southward displacement of Indochina

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Received 9 August 1998; revised version received 2 November 1998; accepted 3 November 1998

Abstract

Middle Cretaceous red sandstones and siltstones were collected at 20 sites for paleomagnetic study from the Nanxin Formation around Yunlong (25.8ºN, 99.4ºE), western part of Yunnan province, China. After stepwise thermal demagnetization, high-temperature component magnetization with unblocking temperature of about 675ºC is isolated. A positive fold test at the 99% confidence limit shows that the high-temperature component magnetization is primary for the middle Cretaceous Nanxin Formation. A tilt-corrected formation mean direction is $D = 40.2º$, $I = 49.9º$ with $\alpha_{95} = 3.9º$, corresponding to a paleopole at 54.6ºN, 171.8ºE with $A_{95} = 4.4º$. Comparison with previously reported Cretaceous paleomagnetic directions indicates that the Yunlong area has been subjected to southward displacement by $11.9 \pm 7.5º$ (corresponding to 1300 ± 800 km) with respect to the stable Yangtze craton since Cretaceous time as well as clockwise rotation through $36.3 \pm 13.6º$. This study elucidates that Indochina was squeezed out of the Asian continent due to collision of India accompanied with clockwise rotation. © 1999 Elsevier Science B.V. All rights reserved.

Keywords: plate collision; deformation; Indochina; Yunnan China; Cretaceous; paleomagnetism

1. Introduction

The India–Asia collision is the most prominent tectonic event of East Asia in Cenozoic time. The Indian continent collided with Asia at about 52 Ma [1] and continued to penetrate into the Asian continent [1,2]. The northward penetration of India gave rise to tectonic deformation in the Asian continent. The deformation feature of the Asian lithosphere due to collision of India with Asia has been studied by both laboratory and numerical experiments [3–7]. Crustal thickening along with latitudinal shortening in front of India and tectonic rotation along both sides are anticipated in the Asian continent.

East Asia consists of three large Precambrian cratons (the Sino–Korea craton, the Tarim craton and the Yangtze craton) and some accretionary terranes (e.g. the Simao Terrane, the Baoshan Terrane, the Qiangtang Terrane and the Lhasa Terrane) [8] (Fig. 1a). Since the Simao Terrane, the northern part of the Indochina block, is situated next to the Indian continent along its east side, the deformation feature of Asia due to the penetration of India is expected to be preserved

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PII: S0012-821X(98)00257-X

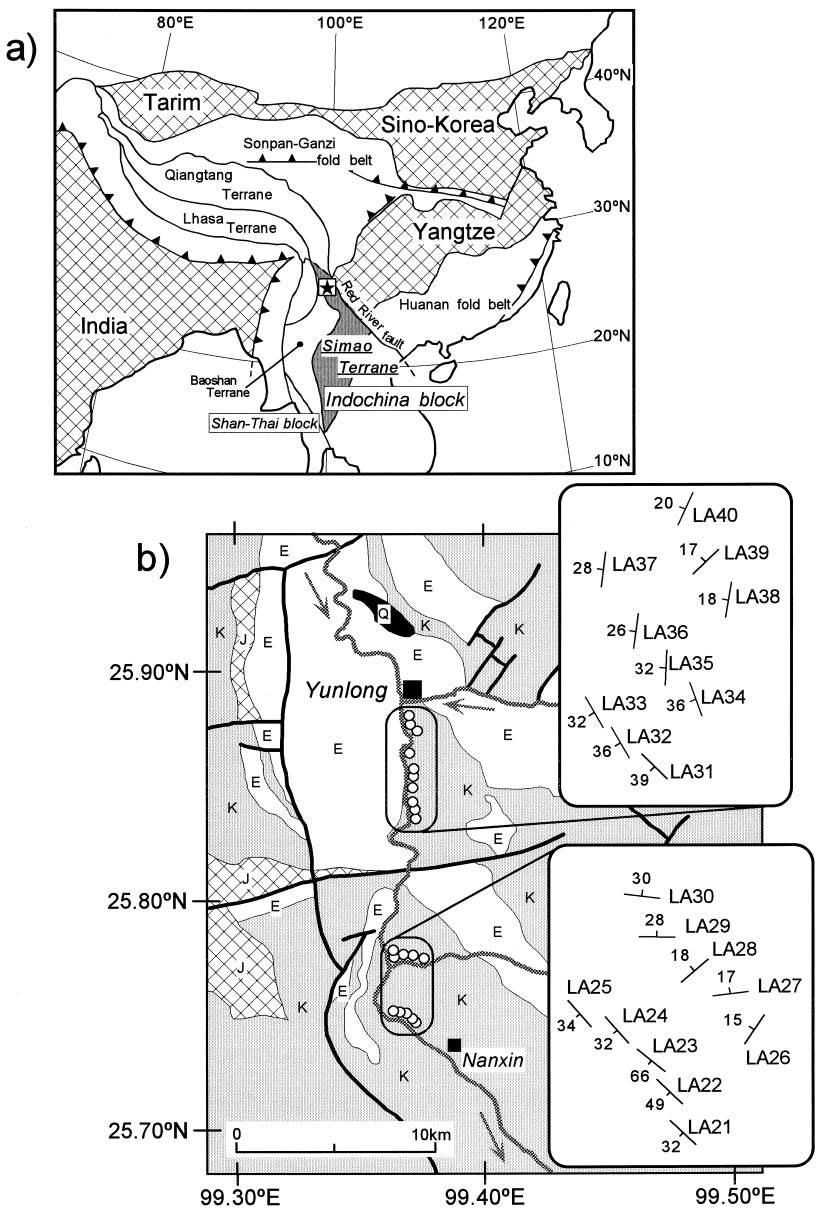


Fig. 1. (a) Sketch tectonic map of Southeast Asia (modified from Zhang et al. [8], Leloup et al. [14] and Yang et al. [34]). The study area is marked with a star. Barbed lines are continental thrusts. (b) Geological map of the study area (after [23,24]). Sampling sites are indicated by circles. Black and gray lines denote faults and rivers, respectively. J = Jurassic; K = Cretaceous; E = Paleogene; Q = Quaternary. Strike and dip of strata for each sampling site are shown in insets.

in the terrane. Paleomagnetic evidence on clockwise tectonic rotation ranging between 42° and 112° has been reported from the Simao Terrane [9–11].

The Yunlong area on the northern part of the Simao Terrane was chosen as a paleomagnetic study area. The present investigation aims at detecting features of tectonic deformation from paleomagnetic view points. Our paleomagnetic study focuses on the middle Cretaceous redbeds of the Nanxin Formation, because late Mesozoic formation records evidence of

Tertiary tectonic deformation due to the penetration of India since about 52 Ma.

2. Geological setting and sampling

The NW–SE-trending Red River fault is a conspicuous tectonic feature of South China. This is the boundary between the Yangtze craton and two accretionary blocks (the Shan-Thai and Indochina blocks). Previous geological, chronological and seismic data [12,14–20] imply that a large amount of sinistral motion took place on the Red River fault in Tertiary time, whereas dextral motion occurred in Quaternary time [13].

The Shan-Thai and Indochina blocks, west of the Red River fault zone, consist of tectonic collages of continental fragments. The Baoshan Terrane and the Simao Terrane are well-defined continental fragments [21] (Fig. 1a). These two terranes are bounded by the Changning–Menglian suture [22,23]. The Baoshan Terrane, a part of the Shan-Thai block to the west of the Changning–Menglian suture, shows very clearly Gondwana affinities [22]. The Simao Terrane, the northwestern part of the Indochina block to the east of the Changning–Menglian suture, is a Cathaysian element [22]. These two terranes have amalgamated to the Yangtze craton until the Late Triassic [22].

Yunlong city (25.8ºN, 99.4ºE) is about 50 km west of the Red River fault and is situated in the Lanping Basin, northern part of the Simao Terrane (Fig. 1b). Around Yunlong, Jurassic to Eocene (locally Quaternary) cover strata are exposed. Oligocene to Pliocene rocks are, however, absent. This hiatus is probably due to Oligo–Miocene décollements in this region [17]. The Cretaceous sequence of terrestrial origin is composed of four formations [24]: the Lower Cretaceous Jinxing Formation, the middle Cretaceous Nanxin Formation, the Upper Cretaceous Hutousi Formation and Mankuanhe Formation, in an ascending order. The middle Cretaceous Nanxin Formation, 1495.5 m in thickness, consists mainly of purplish red sandstones. This formation conformably overlies the Lower Cretaceous Jinxing Formation and conformably underlies the Upper Cretaceous Hutousi Formation. The occurrence of *Estheria*, gastropods, charophytes and sporopollen assigns a middle Cretaceous age to the Nanxin Formation [24].

Red sandstones and siltstones were collected at 20 sites (170 samples) in the Nanxin Formation along a road running parallel to a branch (Yangbijiang) of the Mekong River. The outcrops dip from northward to southeastward at 15º–66º. The folding took place in the Late Eocene as evidenced by an unconformity between the Denghei Formation and the Menglian Formation in the Upper Eocene section [24]. Eight to ten hand samples, oriented with a magnetic compass, were distributed over a distance up to 15 m in a site. Localities of the sampling sites were determined by a portable navigational instrument (SONY PYXIS).

The present declination value at each sampling locality was estimated from the calculation using the International Geomagnetic Reference Field, 1995 [25].

3. Paleomagnetism

In the laboratory, one or more specimens, 25 mm in diameter and about 23 mm in height, were prepared from each sample. Remanent magnetization of each specimen was measured using a 2G Enterprises cryogenic magnetometer. A stepwise thermal demagnetization was carried out up to 690ºC with a laboratory-built furnace or a Natsuhara TDS-1 thermal demagnetizer; the residual field in these furnaces was less than 5 nT. Stepwise alternating field (AF) demagnetization up to 200 mT was also carried out for at least one specimen from each site with a Natsuhara AF demagnetizer. The magnetic behavior for each specimen during demagnetization was plotted on a Zijderveld diagram [26]. Principal component analysis [27] was used to determine the component directions of magnetization, and a site-mean direction was calculated using Fisherian statistics [28].

Progressive acquisition of isothermal remanent magnetization (IRM) up to 2.7 T and thermal demagnetization of composite IRMs [29] were performed on a 2G pulse magnetizer in order to identify ferromagnetic minerals. For composite IRMs, hard, medium and soft components were treated in DC fields of 2.7 T, 0.5 T, 0.15 T, respectively. Thermomagnetic analysis was carried out for several specimens with a recording thermomagnetic balance of Kyoto University.

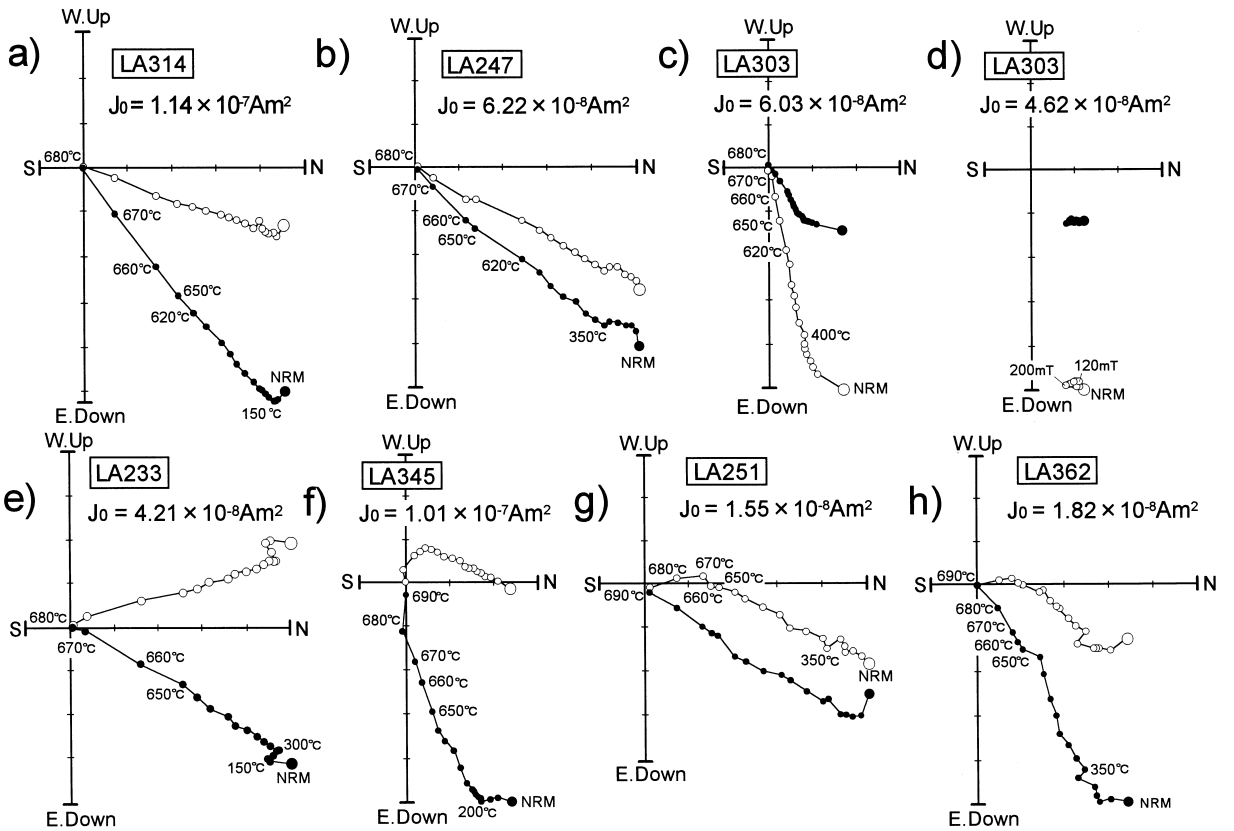


Fig. 2. Orthogonal projections of magnetization vector end-points for representative specimens from the middle Cretaceous Nanxin Formation during thermal and AF demagnetization experiments (in geographic coordinates). Solid symbols for horizontal projection, open symbols for vertical projection.

3.1. Demagnetization

Each specimen has natural remanent magnetization (NRM) intensity of $1.90–0.09 \times 10^{-7}$Am^2, and the majority of the NRM intensities ranges from 0.8 to 0.3×10^{-7}Am^2. All specimens have a multi-component magnetization; generally two-component magnetization appears during progressive thermal demagnetization. The low-temperature component is removed by 300ºC. After removal of the low-temperature component, remanent magnetization decays linearly to the original value (Fig. 2a–c, e). The unblocking temperature of the high-temperature component ranges between 670ºC and 690ºC. Some specimens have middle-temperature component magnetization (Fig. 2f–h). Two specimens (LA251 and LA362 from sites LA25 and LA36, respectively) preserve the middle-temperature component magnetization by 670ºC. AF demagnetization up to 200 mT is not effective to reduce their intensities (Fig. 2d). A high unblocking temperature above 670ºC and high coercivity of more than 200 mT indicate that the remanent magnetization is mainly carried by hematite.

Thermomagnetic analysis clearly shows Curie temperatures of about 675ºC (Fig. 3a). This experiment ensures that the main carrier of remanent magnetization resides in hematite. The IRM acquisition curve of each specimen (Fig. 3b) reveals a ‘shoulder’ at a field of 300 mT. Although the IRM component increases slightly above 300 mT, intensities do not reach the saturation value at 2.7 T. The IRM acquisition experiment shows that each specimen contains at least two (hematite and an unknown) ferromagnetic minerals. Moreover, thermal demagnetization of composite IRMs indicates un-

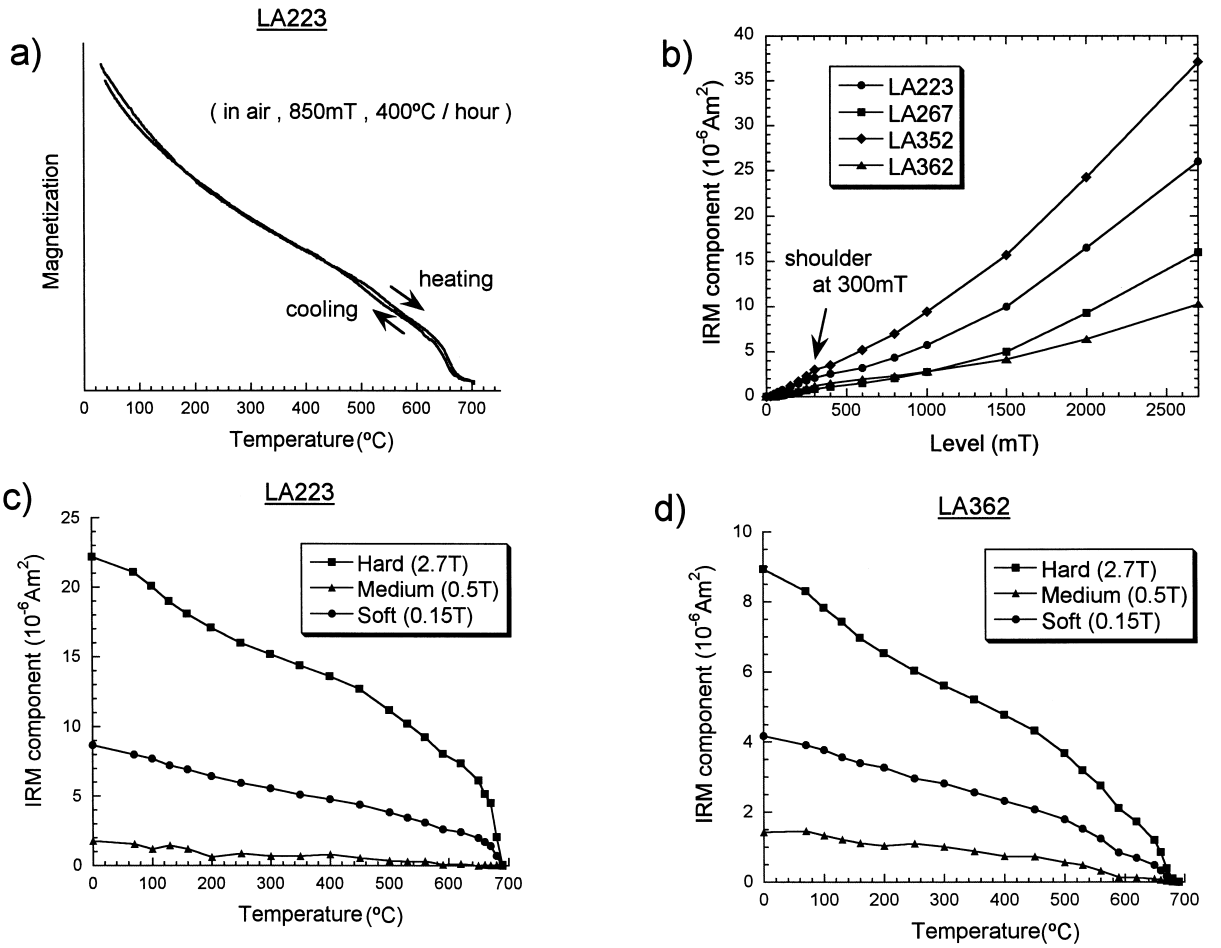


Fig. 3. (a) Thermomagnetic curve for a representative specimen (LA223) showing the Curie point of hematite at about 675ºC. High field magnetization is conventionally in arbitrary units. (b) IRM acquisition curves for the representative specimens LA223, LA267, LA352 and LA362. Each specimen reveals a shoulder at 300 mT and does not reach saturation IRM at the field of 2.7 T. (c, d) Thermal demagnetization of composite IRMs (2.7 T, 0.5 T and 0.12 T in DC field for three perpendicular axes of a sample) for the representative specimens LA223 and LA362.

blocking temperatures of about 150ºC in the hard component and about 580ºC in the soft component (Fig. 3c, d). These facts show that trace amounts of goethite and magnetite are present in the redbeds of the Nanxin Formation.

3.2. Paleomagnetic direction

As a result of stepwise thermal demagnetization, three component (low, middle and high temperature) magnetizations are resolved.

The highest temperature components were isolated from 160 samples. Paleomagnetic directions of the highest temperature component display easterly deflection in declination and are well clustered in each site (Fig. 4a, Table 1); the precision parameter k ranges from 16.5 to 373.5. A mean direction of all sites is $D = 50.9^{\circ}$, $I = 29.8^{\circ}$ with $\alpha_{95} = 11.1^{\circ}$ in situ, and $D = 40.2^{\circ}$, $I = 49.9^{\circ}$ with $\alpha_{95} = 3.9^{\circ}$ after tilt correction. After tilt correction, the precision parameter (k) increases (from 9.6 to 70.0; $k_s/k_g = 7.3$) and passes the fold test at the 99% confidence limit [30]. On applying progressive unfolding with a step of 10% (Fig. 4b), the precision parameter (k) reaches a maximum at 100% unfolding, which strongly suggests that the mean direction after tilt correction is

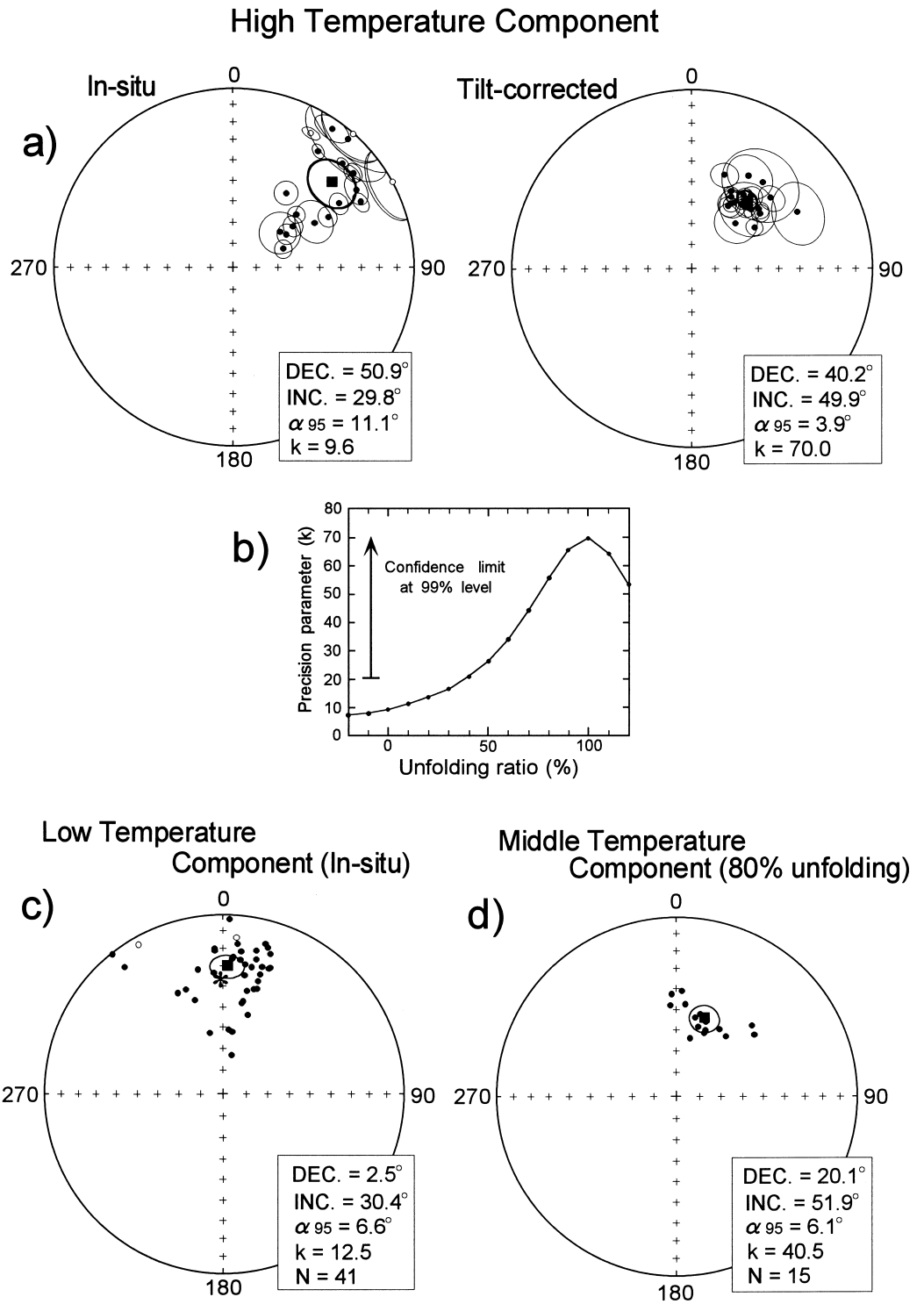

High Temperature Component
a)
In-situ
Tilt-corrected
0
90
180
270
DEC. = 50.9°
INC. = 29.8°
α 95 = 11.1°
k = 9.6
DEC. = 40.2°
INC. = 49.9°
α 95 = 3.9°
k = 70.0
b)
Precision parameter (k)
Confidence limit
at 99% level
Unfolding ratio (%)
Low Temperature
Component (In-situ)
c)
DEC. = 2.5°
INC. = 30.4°
α 95 = 6.6°
k = 12.5
N = 41
Middle Temperature
Component (80% unfolding)
d)
DEC. = 20.1°
INC. = 51.9°
α 95 = 6.1°
k = 40.5
N = 15

Table 1
Paleomagnetic results for redbeds in the middle Cretaceous Nanxin Formation from Yunlong

Site	Locality		Bedding strike/dip	*n*/*N*	In-situ		Tilt-corrected		*k*	α_{95} (°)
	°N	°E			dec. (°)	inc. (°)	dec. (°)	inc. (°)		
High-temperature component:										
LA21	25.75	99.37	129.4/52	5/8	41.7	−0.4	43.0	51.6	40.1	12.2
LA22	25.75	99.37	129.4/49	7/8	35.7	5.8	32.9	54.6	88.4	6.5
LA23	25.75	99.37	124.4/66	10/10	29.9	−15.2	27.6	50.5	264.1	3.0
LA24	25.75	99.37	138.4/32	9/9	36.2	21.6	29.6	52.6	373.5	2.7
LA25	25.75	99.37	138.4134	6/8	41.7	4.9	39.9	38.6	16.5	17.0
LA26	25.78	99.37	213A/15	8/8	52.2	53.2	31.4	55.5	226.2	3.7
LA27	25.78	99.37	266.4/18	8/8	57.3	57.7	39.7	47.1	143.9	4.6
LA28	25.77	99.37	242.4/18	8/8	60.5	62.3	30.6	57.0	163.3	4.3
LA29	25.77	99.36	270.4/28	8/8	72.1	66.3	35.7	48.6	220.6	3.7
LA30	25.78	99.36	276.4/29	6/8	54.5	63.7	31.8	40.0	48.3	9.7
LA31	25.84	99.37	135.4/39	8/8	52.3	17.2	57.1	55.8	242.9	3.6
LA32	25.84	99.37	149.4136	10/10	46.7	18.0	39.3	52.7	321.4	2.7
LA33	25.85	99.37	149.4/32	9/9	51.9	17.2	48.5	48.8	171.0	3.9
LA34	25.85	99.37	159.4/36	8/8	63.1	−1.1	61.9	34.5	19.8	12.8
LA35	25.86	99.37	181.4/32	7/8	64.5	22.8	51.1	49.8	234.5	4.10
LA36	25.86	99.37	186.4/26	10/10	59.3	21.7	48.4	41.1	77.6	5.5
LA37	25.87	99.37	187.4/28	9/9	60.5	33.1	41.1	52.8	128.9	4.6
LA38	25.87	99.37	188.4/18	8/8	62.9	48.1	44.8	61.2	38.0	9.1
LA39	25.88	99.37	224.4117	8/9	37.4	49.5	19.6	44.6	108.1	5.4
LA40	25.88	99.37	202.4120	8/8	63.7	40.8	45.8	51.7	164.4	4.3
Formation mean direction				20/20	50.9	29.8			9.6	11.1
							40.2	49.9	70.0	3.9
Low-temperature component:										
Mean direction				41/170	2.5	30.4			12.5	6.6
							−14.9	35.9	6.0	10.0
Middle-temperature component:										
Mean direction				15/170	32.5	36.2			15.0	10.2
							15.1	55.1	36.1	6.4
(80% unfolding)							20.1	51.9	40.5	6.1

The magnetization directions of high temperature components are listed. *N* and *n* are numbers of field oriented samples and samples used for statistics, respectively. Strikes of bedding are values after subtraction of local geomagnetic declination calculated from the International Geomagnetic Reference Field [25] because directions of strikes were oriented using a magnetic compass. Dec. and inc. are declination and inclination, respectively; *k* is the Fisherian precision parameter [28]; α_{95} is the radius of the cone of 95% confidence about the mean direction. Middle Cretaceous mean directions for high-temperature components are calculated from site mean directions, while mean directions for low- and middle-temperature components are calculated on the basis of sample directions.

characteristic direction of pre-fold origin. We also apply the McFadden's fold test [31]. The calculated value is $\xi_1 = 18.45$ in situ and $\xi_1 = 1.21$ after tilt correction, while the critical value at the 99% confidence level is $\xi = 7.3$ [31]. These indicate that both fold tests are positive at the 99% confidence level. Consequently, the tilt-corrected mean direction of the high-temperature component is the characteristic di-

Fig. 4. (a) Equal-area projections of the site mean directions (circles) of high-temperature components with circles of the 95% confidence limit before and after tilt correction. The formation mean direction is shown by a square. (b) Progressive unfolding of the site mean directions with steps of 10%. The precision parameter *k* reaches a maximum at 100% unfolding. (c) Equal-area projections of low-temperature components before tilt correction. The mean direction is shown by a square. An asterisk represents the present direction of the Earth's magnetic field. (d) Equal-area projections of middle-temperature components at 80% unfolding. The mean direction is shown by a square. The solid symbols refer to the lower hemisphere, the open symbols refer to the upper hemisphere.

rection of the middle Cretaceous Nanxin Formation, which corresponds to a paleomagnetic pole at 54.6ºN, 171.8ºE with $A_{95} = 4.4^{\circ}$.

Low temperature components were resolved by 300ºC from 41 samples (Fig. 4c). The mean direction of this component is $D = 2.5^{\circ}$, $I = 30.4^{\circ}$ with $\alpha_{95} = 6.6^{\circ}$, k = 12.5 in situ, and $D = -14.9^{\circ}$, $I = 35.9^{\circ}$ with $\alpha_{95} = 10.0^{\circ}$, $k = 6.0$ after tilt correction. Progressive unfolding indicates that the precision parameter (k) reaches the maximum at 10% unfolding ($D = 1.5^{\circ}$, $I = 30.9^{\circ}$ with $\alpha_{95} = 6.5^{\circ}$, $k = 12.6$). However, little improvement in clustering is observed when compared with the value of the precision parameter (k) at 0% unfolding (increased value of precision parameter k is only 0.1). The mean direction in situ (0% unfolding) is sub-parallel to the present direction of the Earth's magnetic field ($D = 359.4^{\circ}$, $I = 38.3^{\circ}$). These suggest that this component is probably attributable to viscous remanent magnetization acquired in recent times.

Middle-temperature components were isolated by 650ºC–670ºC only from 15 samples out of 170 samples (Fig. 4d). The middle-temperature components reveal easterly and downward directions. The mean direction is $D = 32.5^{\circ}$, $I = 36.2^{\circ}$ with $\alpha_{95} = 10.2^{\circ}$ in situ and $D = 15.1^{\circ}$, $I = 55.1^{\circ}$ with $\alpha_{95} = 6.4^{\circ}$ after tilt correction. The precision parameter (k) increases from 15.0 to 36.1 ($k_s/k_g = 2.41$) by tilt correction. However, while applying progressive unfolding, the precision parameter (k) reaches a maximum at 80% unfolding ($D = 20.1^{\circ}$, $I = 51.9^{\circ}$ with $\alpha_{95} = 6.1^{\circ}$, $k_{80\%} = 40.5$; $k_{80\%}/k_g = 2.7$). The folding took place in the Late Eocene [24], so that this remagnetization was acquired probably in the Late Eocene. We note the Tertiary paleomagnetic result reported by Chen et al. [11]. They recognized their Paleocene data as a remagnetization acquired during the Miocene because their fold test is inconclusive. Their Paleogene data in situ ($D = 42.8^{\circ}$, $I = 26.1^{\circ}$ with $\alpha_{95} = 10.3^{\circ}$) is, however, significantly different from their Miocene data ($D = 21.1^{\circ}$, $I = 35.5^{\circ}$ with $\alpha_{95} = 7.1^{\circ}$). Our middle-temperature component unfolded at the 80% level ($D = 20.1^{\circ}$, $I = 51.9^{\circ}$ with $\alpha_{95} = 6.1^{\circ}$) is similar to that of the tilt-corrected Paleocene data of Chen et al. [11] ($D = 23.9^{\circ}$, $I = 51.6^{\circ}$ with $\alpha_{95} = 7.8^{\circ}$), which implies that our middle-temperature component and their Paleocene directions were probably formed during an earlier stage of folding in the Eocene. However, the number of our samples is not enough to give a certain conclusion.

4. Discussion

The middle Cretaceous Nanxin Formation in the Yunlong area preserves a highly reliable paleomagnetic direction: an easterly deflected declination is observed ($D = 40.2^{\circ}$, $I = 49.9^{\circ}$ with $\alpha_{95} = 3.9^{\circ}$). The reliability of the primary nature for the remanent magnetization is ascertained through successful thermal demagnetization and a positive fold test. Characteristic directions for all sites after tilt correction show normal polarity, indicating that the characteristic remanent magnetization carried by hematite was acquired during the Cretaceous Normal Superchron [32,33].

4.1. Paleomagnetic direction of the northern part of the Simao Terrane

The clockwise deflected declination of Yunlong ($D = 40.2^{\circ}$) is comparable with those previously reported from the northern part of the Simao Terrane. Easterly deflected declination ($D = 42.0^{\circ}$) was reported by Funahara et al. [9] near Yongping (25.5ºN, 99.5ºE). The primary nature of the magnetization was ascertained through a positive fold test at the 99% confidence level, although the mean direction of twelve sites has a large α_{95} of 15.7º. This somewhat large uncertainty is related to a large dispersion of the declinations. It is probably ascribed to local vertical axis rotations of the sites or to a lack of structure correction (e.g. tilt of plunge). Our results reconfirm the clockwise rotation of the paleomagnetic direction at Yongping [9]. Our reliable paleomagnetic direction of Yunlong is therefore recognized as representative of the northern part of the Simao Terrane.

A northerly direction ($D = 6.9^{\circ}$) is reported for the middle Cretaceous Nanxin Formation from Xiaguan (25.6ºN, 100.2ºE) [10]. Its reliability was established by a positive fold test along with a positive reversal test. Since clockwise rotations of about 40º are observed from the Yunlong area (25.8ºN) through the Yongping area (25.5ºN), the paleomagnetic data of Xiaguan are enigmatic. One possible explanation for

the northerly direction is a wrong age assignment to the magnetization [10,34]. An alternative explanation is local block rotation of the Xiaguan area due to dextral strike-slip motion of the Red River fault in Pliocene time [13]. The northerly direction is probably a local phenomenon specific to Xiaguan and is hardly to accept as the Cretaceous characteristic direction for the northern part of the Simao Terrane.

4.2. Comparison with other Cretaceous paleomagnetic data obtained around the Red River fault

4.2.1. Comparison with Cretaceous paleomagnetic data of the southwestern part of the Yangtze craton

The Yunlong pole is compared with the Cretaceous paleomagnetic poles of Ya'an (30.0ºN, 102.9ºE) [35], Xichang (27.9ºN, 102.3ºE) [36] and Huili (26.6ºN, 102.3ºE) [37] from the southwestern part of the Yangtze craton (Fig. 5a). These three results are reliable because of successful thermal demagnetization and positive fold tests. The poles of the Yangtze craton are indistinguishable from one another at the 95% confidence level, so that these three poles are averaged (82.2ºN, 250.3ºE with $A_{95} = 7.6^{o}$). We recognize this mean pole as a Cretaceous characteristic pole of the stable Yangtze craton (Table 2).

The Yunlong pole is significantly different from the mean pole of the stable Yangtze craton (Fig. 5a). Angular distance between the Yunlong pole and the mean pole of the stable Yangtze craton is 34.7 ± 8.8^{o}. This indicates that the Yunlong area was subjected to tectonic movement with respect to the stable

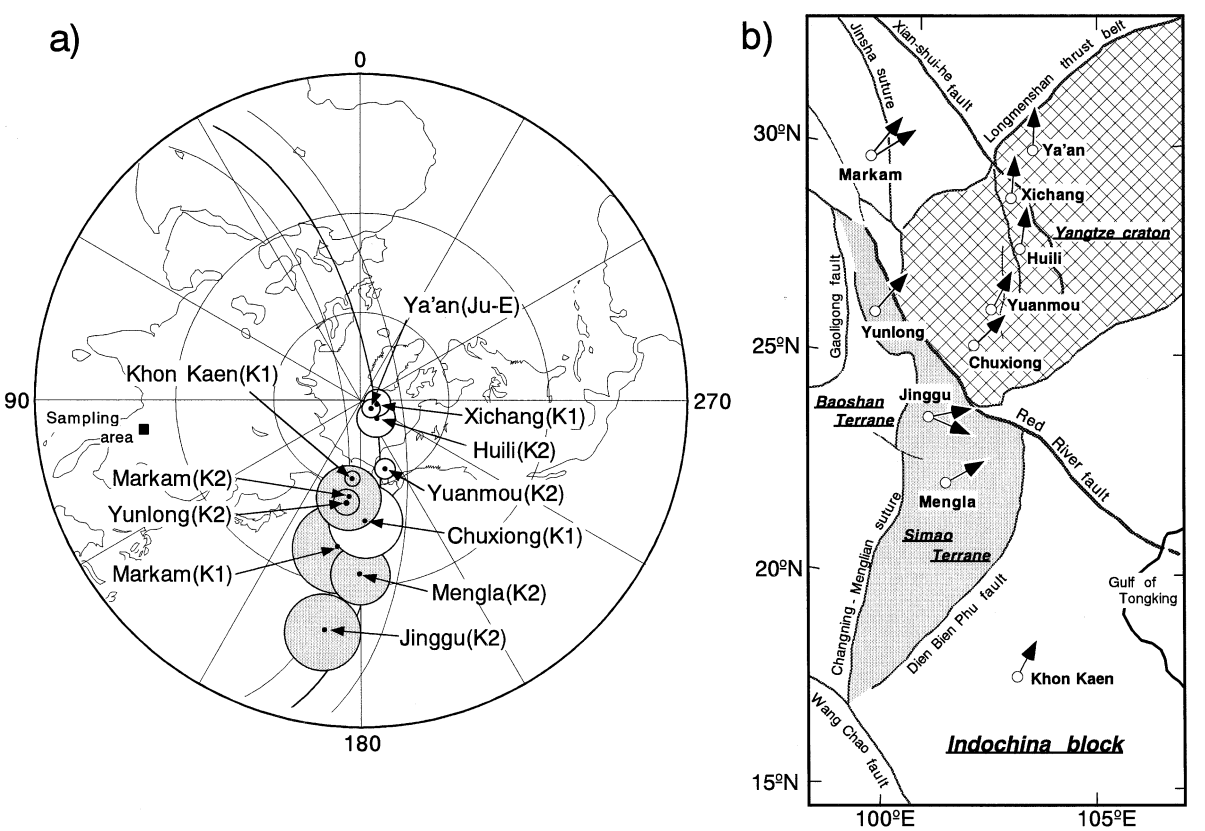


Fig. 5. (a) Cretaceous paleomagnetic poles of eleven areas around the Red River fault with associated circles of 95% confidence. Small circles centred in our study area are also shown (see text). The shaded poles are from west of the Red River fault (the Qiangtang Terrane and the Indochina block) and the unshaded poles are from southwestern part of the Yangtze craton. *Ju* = Upper Jurassic; *K1* = Lower Cretaceous; *K2* = Upper Cretaceous; *E* = Paleogene. (b) Declinations of the Cretaceous paleomagnetic directions of the Indochina block and its surrounding regions are indicated by arrows.

Table 2
Cretaceous paleomagnetic results obtained from the Indochina block and its surrounding regions

Locality			Age	*N*	Observed direction			VGP			Rotation (°)	Latitudinal difference (°)	Reference
name	lat. (°N)	lon. (°E)			dec. (°)	inc. (°)	α_{95} (°)	lat. (°N)	lon. (°E)	A_{95} (°)			
Indochina block													
Simao Terrane:													
Yunlong	25.8	99.4	K2	20	40.2	49.9	3.9	54.6	171.3	4.4	36.3 ± 13.6	11.9 ± 7.5	This study
Xiaguan	25.6	100.2	K2	9	6.9	47.7	8.6	83.6	152.7	10.0	2.9 ± 17.7	10.2 ± 9.9	[10]
Yongping	25.5	99.5	K1	12	42.0	51.1	15.7	50.9	167.3	20.6	38.1 ± 28.3	12.0 ± 9.4 *	[9]
Jinggu	23.5	100.6	K1·K2	–	112.6	34.7	7.5	−11.7	163.6	–	108.5 ± 15.4	2.5 ± 8.3	[11]
Jinggu	23.4	100.9	K2	8	79.4	43.3	9.1	18.9	170.0	8.9	75.3 ± 17.6	8.7 ± 9.7	[10]
Mengla	21.6	100.4	K2	10	60.8	37.8	7.6	33.7	179.3	8.2	56.8 ± 15.7	6.5 ± 8.5	[10]
Khorat plateau:													
Khon Kaen	16.5	103.0	K1	10	28.1	40.5	2.4	62.7	173.3	2.4	23.8 ± 13.2	13.3 ± 6.9	[43]
Yangtze craton													
Ya'an	30.0	102.9	Ju–E	26	2.0	34.2	3.6	78.6	273.4	2.9			[35]
Xichang	27.9	102.3	K1	36	3.4	41.2	5.3	84.8	245.8	5.7			[36]
Huili	26.6	102.3	K2	18	8.1	38.8	6.6	81.5	220.9	7.1			[37]
(stable Yangtze craton)								82.2	250.3	7.6			
Yuanmou	25.9	101.8	K2	21	26.9	35.6	3.6	64.6	199.6	3.3			[38]
Chuxiong	25.0	101.5	K1	21	44.6	41.3	10.7	49.2	182.0	11.4			[39]
Qiangtang Terrane)													
Markam	29.7	98.7	K2	11	39.7	51.6	8.5	56.7	172.7	10.6	35.8 ± 18.2	8.4 ± 11.9	[42]
Markam	29.7	98.4	K1	12	57.7	50.2	10.9	40.6	170.5	13.0	53.8 ± 20.9	9.6 ± 10.4	[42]

Lat. and lon. are latitude and longitude, respectively. Age: Ju = Upper Jurassic; K1 = Lower Cretaceous; K2 = Upper Cretaceous; E = Paleogene. *N* is number of sites used for statistics. Dec. and inc. are declination and inclination, respectively. α_{95} and A_{95} are the radius of the cone of 95% confidence about the mean direction and the VGP, respectively. Rotation and latitudinal differences are evaluated by comparison between the observed paleomagnetic declination and that calculated from the stable Yangtze craton. Latitudinal difference of Yongping (*) is evaluated using recalculated inclination data (bivariate mean) by Yang et al. [34].

Yangtze craton. This deviation can not be explained by the wandering of the paleomagnetic pole because the Eurasian apparent polar wander path during the Cretaceous is roughly quasi-stationary [40,41].

4.2.2. Clockwise rotation of the Indochina block

One of the prominent tectonic phenomena for the Yunlong area is the clockwise rotation with respect to the stable Yangtze craton. The amount of rotation is evaluated from comparison between observed and expected paleomagnetic directions calculated from the pole of the stable Yangtze craton at Yunlong: the Yunlong area is rotated clockwise through $36.3 \pm 13.6^{\circ}$.

The clockwise tectonic rotation of 25°–40° is also observed in terranes along the Yangtze craton: the eastern Qiangtang Terrrane and the Simao Terrane (Fig. 5b). Paleomagnetic data from Markam (30°N, 99°E) in the eastern Qiangtang Terrane [42] show clockwise rotations of about 36°–54° with respect to the stable Yangtze craton. A clockwise tectonic rotation of more than 57° is also expected in the Simao Treeane [10,11]. Easterly declinations are reported from Jinggu (23.4°N, 100.9°E; $D = 79^{\circ}$) and Mengla (21.6°N, 100.4°E; $D = 61^{\circ}$) of the middle Cretaceous Mangang Formation in the central part of the Simao Terrane [10]. Paleomagnetic results of the Simao region (Jinggu; 23.5°N, 100.6°E) from Cretaceous redbeds have been also reported by Chen et al. [11]. Their paleomagnetic directions of the Mangang Formation ($D = 84^{\circ}$) are consistent with the paleomagnetic data from Jinggu [10], though the fold tests they demonstrated are based on the sample level. Paleomagnetic data near Khon Kaen (Khorat plateau; 16.5°N, 103.0°E) in the central part of the Indochina block [43] show a clockwise rotation of $23.8 \pm 13.2^{\circ}$ with respect to the stable Yangtze craton. Deformation of terranes to the west of the Yangtze craton is characterized by a clockwise rotation of more than 30°.

The deformed zone extends 1000 km along the Yangtze craton: from Markam (30°N, 99°E) to Khon Kaen (16°N, 103°E). The paleomagnetic poles obtained from the Indochina block, west of the Red River fault, are streaked out along longitude 170°E and they are distributed along a small circle centred in our study area (Fig. 5a). This indicates that differential rotations took place for these areas. Since the existence of a fairly long deformation zone is expected northeast of India from the laboratory experiments of Tapponnier et al. [3], the clockwise rotation observed in the deformed zone is ascribed to tectonic deformation due to the collision of India with Asia. The deformed zone experienced a variable rotation rather than a uniform rotation. The rotation of the Jinggu and Mengla areas in the Simao region, the central part of the Simao Basin in the Simao Terrane, is considerably larger (more than 30°) than that of the northern part of the Simao Terrane or central part of the Khorat plateau. This implies that the Indochina block has not behaved as a coherent unit block since Cretaceous time. Differential rotation is not attributed to local rotation due to fault movements, because no prominent fault zones or shear zones are present in the Simao region. The large and variable clockwise rotation observed in the Simao region is rather ascribed to deformation of the soft and deformable Indochina block which flanks the stiff Yangtze craton. The large rotation due to regional deformation of the Indochina block is a more preferable explanation than local deformation associated with motion of strike-slip faults such as the plane-strain 'domino' model [44–46].

4.2.3. Southward extrusion of Indochina

Latitudinal displacement is claimed for the terranes west of the Yangtze craton from comparison of the paleomagnetic poles. The amount of southward displacement is estimated for the Yunlong area with respect to the stable Yangtze craton. The paleolatitude of the Yunlong area calculated from the observed inclination is $30.7 \pm 3.5^{\circ}$N, whereas the paleolatitude of the Yunlong area expected from the stable Yangtze craton is $18.8 \pm 6.7^{\circ}$N. The amount of latitudinal difference between the observed paleolatitude and the expected paleolatitude is $11.9 \pm 7.5^{\circ}$, corresponding to 1300 ± 800 km (Fig. 6).

Southward displacement more than 1000 km has been inferred from the Indochina block [34,43]. Yang and Besse [43] suggest $11.5 \pm 6.7^{\circ}$ of southward displacement of the Indochina block and Yang et al. [34] suggest $8.4 \pm 3.0^{\circ}$ of southward displacement of the Shan-Thai and Indochina blocks. This southward displacement is probably associated with southeastward extrusion of Indochina along the Red River fault due to indentation of India into Asia

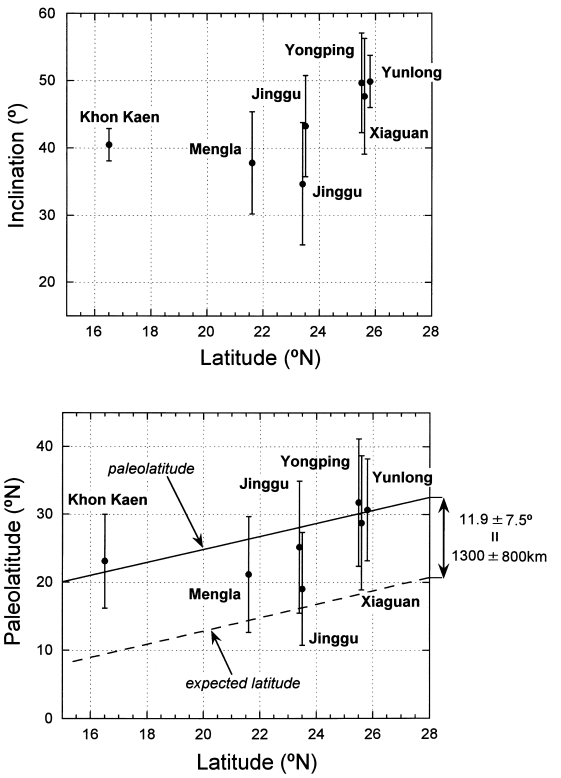


Fig. 6. Plots of inclination and paleolatitude of Yunlong (this study), Xiaguan [10], Yongping [9], Jinggu [10,11], Mengla [10] and Khon Kaen [43] for the Cretaceous. Paleolatitude and its uncertainty are calculated after Butler [51]. The error bar shows uncertainty at the 95% confidence level.

(e.g. [12,14–18,34]). Geological and chronological data show that 700 ± 200 km of sinistral motion along the Red River fault occurred between 35 Ma and 17 Ma [16,19,47]. The estimated amount of the southeastward motion from a paleomagnetic view point is much greater (although compatible within the 95% confidence limit) than the estimated amount of the southeastward displacement from geological and chronological data.

The Qiangtang Terrane north of the Simao Terrane, however, experienced no southward displacement. Paleomagnetic data from Markam [42] suggest insignificant latitudinal displacement (8.4 ± 11.9° southward displacement for the Lower Cretaceous Cuowa Formation, and 9.6 ± 10.4° southward displacement for the middle Cretaceous Mankang Formation). Since the Red River fault does not extend northward beyond the Jiali–Parlung fault, not any terrane north of the Jiali–Parlung fault has moved southward. Tectonic deformation in eastern Qiangtang Terrane was accommodated only by clockwise rotation.

4.3. Synthetic tectonic model

A realistic tectonic model is synthesized on the basis of paleomagnetic, geological and chronological evidence (Fig. 7).

(A) After India collided with Asia in the Early Eocene, India began to penetrate into Asia.

(B) Tectonic deformation of the Asian continent commenced since the Late Eocene. The penetration of India was accommodated by crustal thickening in front of India: the Tibetan plateau and the Himalayas [48]. In the Indochina block east of India, the penetration of India was accommodated by a clockwise rotational motion and by a southward displacement of about 1300 km. This large displacement was associated with left-lateral motion of the Red River fault. The Qiangtang Terrane was only subjected to clockwise rotation.

(C) The deformation of the Asian continent continued until Miocene to Pliocene times. The Miocene penetration of India was absorbed by thrusting in the Himalayas and the Longmen Mountains which started at around 20 Ma [16,49,50]. The Simao region, in the central part of the Simao Basin in the Simao Terrane, might suffer additional clockwise rotation of 20°–40° in this time. This rotational mode is probably associated with the eastward décollements which occurred in this region during the Miocene [17].

Acknowledgements

The thermomagnetic analysis was made possible by courtesy of Masayuki Torii of the Division of Earth and Planetary Science, Kyoto University. We thank R.J. Enkin and one anonymous reviewer for the critical reviews of the manuscript. We are also grateful to B.C. Huang for useful discussions. The research was partly supported by a Grant-in-aid (No. 060401079; 09041109) from the Japanese Ministry of Education, Science, Sports and Culture. ***[RV]***

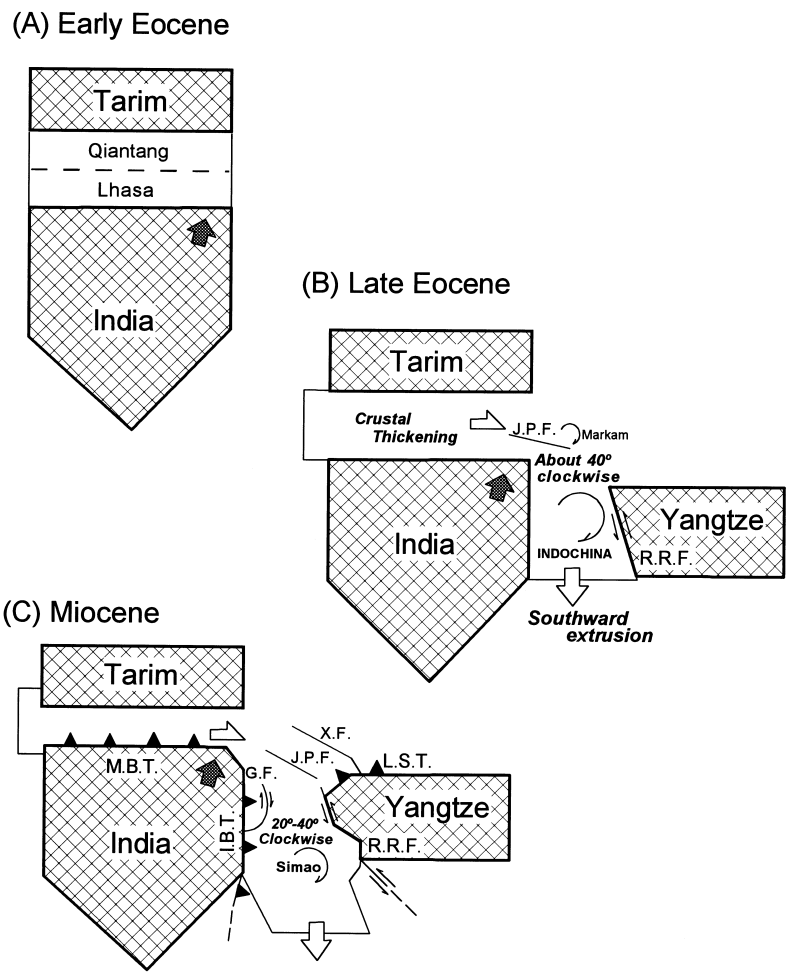


Fig. 7. A synthetic tectonic model on the basis of paleomagnetic, geological and chronological evidence. (A) After India collided with Asia in the Early Eocene, India began to penetrate into Asia. Shaded arrow indicates direction of motion of the Indian continent relative to Eurasia. (B) Tectonic deformation of the Asian continent commenced since the Late Eocene. The penetration of India was accommodated by crustal thickening in front of India. In the Indochina block, east of India, the penetration of India was accommodated by southward displacement of about 1300 km and by clockwise rotational motion. This large displacement was associated with left-lateral motion of the Red River fault (*R.R.F.*). The eastern part of the Qiangtang Terrane (Markam) north of the Jiali–Parlung fault (*J.P.F.*) was only subjected to clockwise rotation. Open arrows indicate direction of displacement of terranes. (C) Deformation of the Asian continent has continued until the Miocene to Pliocene. Thrusting at the following thrusts started at around 20 Ma [16,49,50]: the Main Boundary thrust (*M.B.T.*) of the Himalayas, the Indo–Burman thrust (*I.B.T.*) of the Indo–Burman Ranges and the Longmenshan thrust (*L.S.T.*) of the Longmen Mountains. The Miocene penetration of India was absorbed by these thrusts. The Simao region, central part of the Simao basin in the Simao Terrane, might suffer an additional clockwise rotation of 20°–40° in this time. *G.F.* = Gaoligong fault; *X.F.* = Xian-shui-he fault.

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